Safaeid Hossain Arib

safaeid48@gmail.com · +880 1677 850587

Website · Google Scholar · GitHub · LinkedIn

Education

University of Dhaka Jan 2019 – Mar 2024
BSc in Robotics and Mechatronics Engineering Dhaka, Bangladesh
CGPA: 3.66/4.00 (Final year: 3.77/4.00)

Selected coursework: Artificial Intelligence; Introduction to Machine Learning; Digital Image Processing and Robot Vision; Digital Signal Processing; Object-Oriented Programming; Fundamentals of Programming; Linear Algebra; Multivariate and Vector Calculus; Statistics for Engineers.

Thesis: Continuous Bangla Sign Language Translation: Mitigating the Expense of Gloss Annotation with the Assistance of Graph. Thesis book

Research Focus

I study how intelligent agents can learn from experience and multimodal feedback to reason and act efficiently. My research spans reinforcement learning, world models, on-policy distillation, multimodal learning, and language model reasoning, with computational efficiency and robust generalization as unifying goals.

Publications & Preprints

[1] **Teach Yourself Where to Look: On-Policy Attention Self-Distillation for Reasoning**
S. H. Arib, R. Akter, I. N. Swapnil, Md F. A. Sayeedi, T. Mohiuddin, Md M. Islam
On-policy attention self-distillation improves mathematical reasoning accuracy by 4.98–8.40 percentage points.
Under review at ICLR 2027 · Paper

[2] **LAGEA: Language Guided Embodied Agents for Robotic Manipulation**
A. M. Chowdhury, A. M. R. Mazumder, **S. H. Arib**, R. Akter
Uses vision-language feedback on robot failures to shape reinforcement learning rewards for manipulation.
ICML 2026 · Paper

[3] **T3Time: Tri-Modal Time Series Forecasting via Adaptive Multi-Head Alignment and Residual Fusion**
A. M. Chowdhury, R. Akter, **S. H. Arib**
Aligns time, frequency, and language features for multivariate time series forecasting; 3.28% lower mean squared error.
AAAI 2026 · Paper

[4] **Counting Through Occlusion: Framework for Open World Amodal Counting**
S. H. Arib, R. Akter, A. M. Chowdhury, Md J. A. Sourov, Md M. Hasan
Reconstructs occluded object features and aligns visual attention for robust open-world amodal counting.
Under review at WACV 2027 · Paper

[5] **SignFormer-GCN: Continuous Sign Language Translation Using Spatio-Temporal Graph Convolutional Networks**
S. H. Arib*, R. Akter*, S. Rahman, S. Rahman (*equal contribution)
Combines video and skeletal features for gloss-free translation of Bangla, American, and German sign languages.
PLOS ONE · WiML Workshop, NeurIPS 2025 · Paper · Extended abstract

[6] **Bornil: An open-source sign language data crowdsourcing platform for AI enabled dialect-agnostic communication**
S. E. Dhruvo, M. A. Rahman, M. K. Mandal, Md I. H. Shihab, A. A. N. Ansary, K. F. Shithi, S. Khanom, R. Akter, **S. H. Arib**
Enables multilingual sign language data collection, annotation, and validation for low-resource recognition.
Preprint · Paper

[7] **Classical machine learning approach for human activity recognition using location data**
S. H. Arib, R. Akter, O. Shahid, Md A. R. Ahad
Recognizes eight locomotion and transport activities from smartphone location features using random forests.
Adjunct Proceedings of UbiComp / ISWC 2021 · Paper · Poster

Research Experience

Research Collaboration Mar 2026 – Present
Collaborators: Dr. Md Mofijul Islam (Amazon GenAI, USA); Dr. Tasnim Mohiuddin (QCRI, Qatar) Remote

- Co-developed solution-conditioned attention supervision for **on-policy self-distillation**; improved Qwen3 mathematical reasoning by **4.98–8.40 percentage points** over token-only self-distillation with **72.6% less estimated model computation**.

Research Collaboration Sep 2025 – Present
Collaborator: Dr. Ser-Nam Lim (University of Central Florida, USA) Remote

- Working on **world models** for reinforcement learning and **on-policy distillation** for language model reasoning.

University of Dhaka Jun 2025 – Present
Research Assistant Dhaka, Bangladesh
Department of Robotics and Mechatronics Engineering

Supervisor: Dr. Md Mehedi Hasan

- Developed **CountOCC**, reconstructing occluded object features with multimodal guidance and attention consistency; reduced mean absolute counting error by **20.80%** on FSC-147-OCC test and **49.89%** on CARPK-OCC over prior baselines.

University of Dhaka Jan 2023 – Dec 2023
Undergraduate Research Assistant Dhaka, Bangladesh
Department of Robotics and Mechatronics Engineering

Supervisors: Dr. Shafin Rahman and Dr. Sejuti Rahman

- Developed **SignFormer-GCN**, fusing video and skeletal representations for gloss-free sign language translation; achieved **19.75 BLEU-4** on RWTH-PHOENIX-2014T and **8.53 BLEU-4** on How2Sign test with **9.43 million parameters**.

Professional Experience

ACI PLC Sep 2025 – Present
Machine Learning Engineer Dhaka, Bangladesh

- Developed few-shot product classification and improved truck identification through license plate enhancement and optical character recognition.
- Building a lightweight text-to-speech model with few-shot voice cloning and adaptation to low-resource languages.
- Developing an agentic assistant for meeting scheduling, document summarization, and conversational file search through Drive and Calendar integration.

Ontik Technology Feb 2024 – Aug 2025
AI Engineer Dhaka, Bangladesh

- Developed agentic B2B automation systems for social media insights and client-ready reporting, substantially reducing manual workflows.
- Built agentic ETL pipelines for COBOL copybook parsing and legacy modernization, enabling AI-driven analytics; demonstrated the system to consulting firms including Oliver Wyman.
- Developed a patient-facing OCR tool to scan prescriptions, schedule medication doses, and recommend suitable alternative medicines under Dr. Javed Mostafa.

Teaching Experience

Bangladesh Artificial Intelligence Olympiad Jun 2024 – Aug 2024
Instructor, National Camp Bangladesh

- Designed and delivered lessons on classification algorithms and evaluation metrics for high school students; created quizzes to assess their understanding.

Selected Academic Projects

Automated Stock Trading Code — Aug 2022 – Nov 2022
Supervisor: Dr. Sejuti Rahman · *Reinforcement Learning*

Human Activity Recognition Using Location Data Code — Apr 2021 – Jun 2021
Supervisor: Dr. Md Atiqur Rahman Ahad · *Machine Learning*

Comparative Analysis of Emotional Support Delivery by NAO Robot and Humans Across Varied Emotional States Code · Demo — Aug 2023 – Nov 2023
Supervisor: Dr. Lafifa Jamal · *Human–Robot Interaction*

Technical Skills

Methods	Reinforcement learning, on-policy self-distillation, multimodal learning, language model post-training, computer vision, time series forecasting
Programming	Python
ML frameworks	PyTorch, Scikit-learn
Agents & tools	LangChain, FastAPI, Docker, Git
Data processing	Pandas, NumPy, OpenCV, SciPy

Honors & Academic Activities

- **Reviewer**, AAAI 2026; ICLR 2027
- **Special Innovation Fund Research Grant**, Ministry of Science and Technology (SRG-232431) — 2023–2024
- **Honourable Mention**, NASA Earth Observatory Dashboard Hackathon — 2021
- **11th place**, Sussex Huawei Locomotion Challenge — 2021
- **Student Activity Secretary**, IEEE Robotics and Automation Society, University of Dhaka — 2023
- **Mentor**, Bangladesh Robot Olympiad (BdRO); school robotics workshops — 2019–2020

References

Dr. Shafin Rahman
Associate Professor, ECE, NSU
Undergraduate Thesis Supervisor
shafin.rahman@northsouth.edu
Website · Google Scholar

Dr. Md Mehedi Hasan
Assistant Professor, RME, DU
Research Supervisor
mmhasan.unsw@gmail.com
Website · Google Scholar

Dr. Md Mofijul Islam
Senior Applied Scientist, AWS GenAI
Research Collaborator
akash.cse.du@gmail.com
Website · Google Scholar